Research Portfolio

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CECILIA ANDREA CALLEJAS PASTOR

세씰리아 칼레하스

Biomedical engineer specializing in AI-driven algorithm and software medical technologies. Passionate about applying cutting-edge technology to healthcare and wellness devices. Committed to continuous learning and contributing to the advancement of healthcare technology worldwide!

EXPERIENCE

Present Research Professor at Chungnam National University Hospital
2025 Postdoctoral Researcher at Seoul National University
2024 Postdoctoral Researcher at Chungnam National University
2024 Senior Researcher Intern at MedInTech Inc.

EDUCATION

2024 PhD in Biomedical Engineering, Chungnam National University
2021 MSc in Biomedical Engineering, Chungnam National University
2016 BSc in Biomedical Engineering, Bolivian Catholic University

KEY SKILLS

Languages: Spanish (Native), English (TOEIC 970), Korean (TOPIK 5), Italian
Software: Python, MATLAB, Android Studio/Java, Flutter/Dart, Jira Cloud, REST API
Hardware: SolidWorks, OrCAD, PADS
Firmware: Segger Embedded Studio, STM32 CubeMX

PUBLICATIONS

Journal Articles: 7 (4 published, 3 under review)
Conference Presentations: 9
Patents: 5 (3 completed, 2 under process)

KEY ACHIEVEMENTS

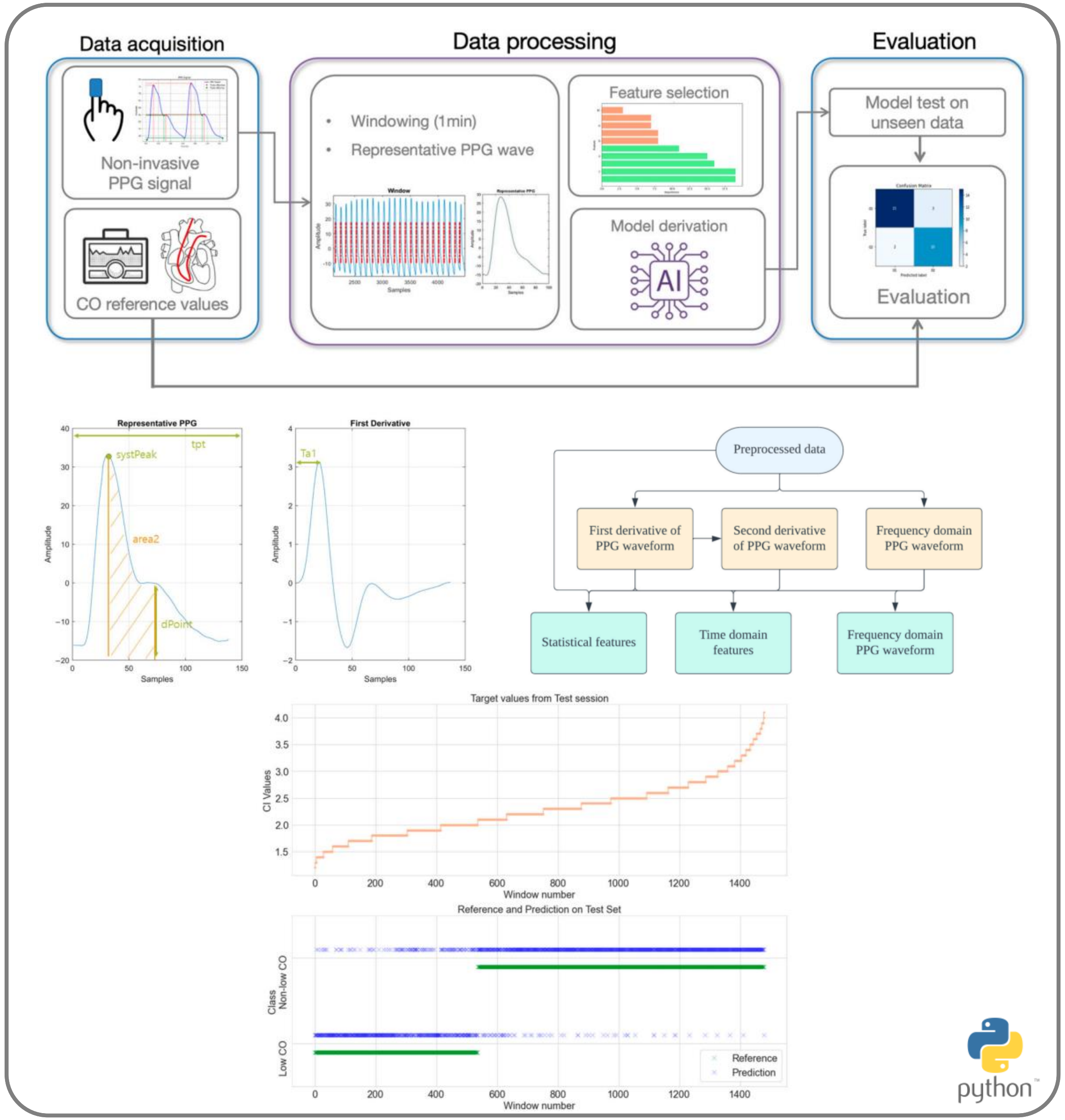
2023 Best Poster Award - Korean Balance Society
2024 Best Poster Award - Korean Innovative Medical Technology Society
2023 Best Oration Award - Korean Balance Society
2022 AI & BigData Scholar - Daewoong Foundation (2022-2024)
2021 Global Korean Scholarship Program Scholar - PhD Program (2021-2024)
2021 AI & BigData Hackathon - Third Place
2021 Best Research Award - Chungnam National University
2018 Global Korean Scholarship Program Scholar - Masters Program (2018-2024)

Contents

- Research in healthcare applications based on physiological signal analysis
- Research on AI-based diagnostic tools
- Research on healthcare wearable system

Research in healthcare applications based on physiological signal analysis

PPG-Based Cardiac Output Estimation

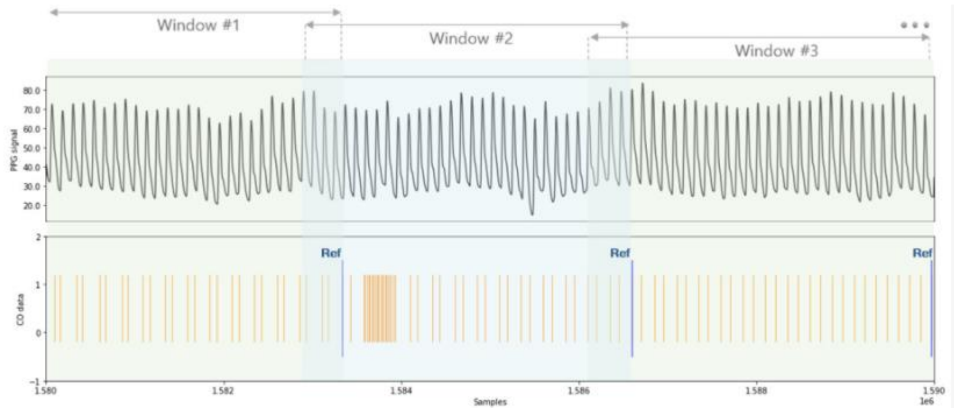


MACHINE LEARNING-BASED CARDIAC INDEX ESTIMATION USING PHOTOPLETHYSMOGRAPHY

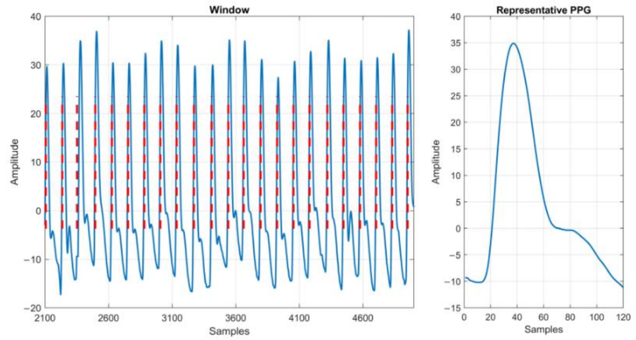
- **Objective:** To develop a non-invasive algorithm to classify cardiac index (CI) using finger photoplethysmography (PPG) and machine learning as an alternative to invasive thermodilution methods.
- **Methods:** The system aimed to accurately identify low versus non-low cardiac output states in patients undergoing off-pump coronary artery bypass surgery
  - Developed non-invasive cardiac output algorithm for classification
  - Used Relief algorithm for optimal feature selection
- **Results:**
  - Implemented CatBoost model achieving 87.57% test accuracy
  - Published in Journal of Clinical Medicine (2024, SCIE Q1)
- **Skills developed:** Feature engineering, ML model selection and optimization, time-series signal processing, physiological data analytics

PPG-Based Cardiac Output Estimation

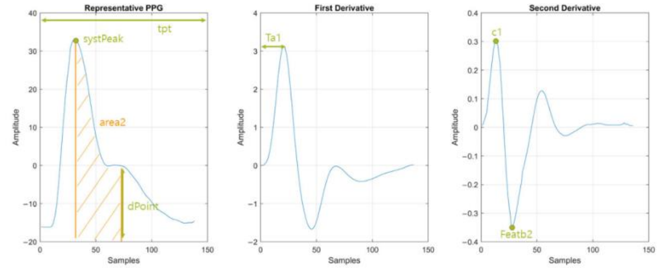
- **Reference Standard:** Used pulmonary artery catheter thermodilution as gold standard for comparison of 67 patients undergoing off-pump coronary artery bypass surgery
- **PPG Signal Processing:**
  - Collected finger PPG at 125 Hz from patients during cardiac surgery
  - Applied bandpass filtering (0.1-15 Hz) and detrending
  - Created representative waveforms by averaging cardiac cycles
- **Feature Engineering:**
  - Extracted 20 key features from PPG waveform and its derivatives
  - Analyzed systolic peaks, dicrotic notches, and spectral components
  - Used Relief algorithm for optimal feature selection
- **Classification Framework:**
  - Categorized cardiac index into "Low CO" (≤2 L/min/m²) vs. "Non-low CO“
  - Implemented CatBoost model with 5-fold cross-validation
  - Validated against gold-standard thermodilution measurements



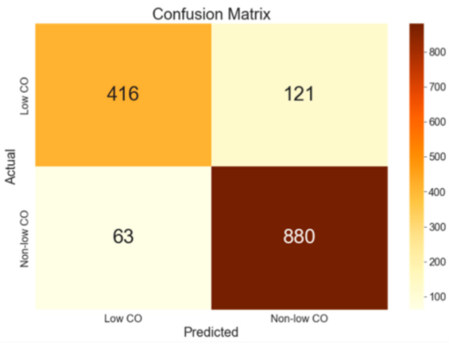
[Windowing processing]



[PPG Signal Processing]

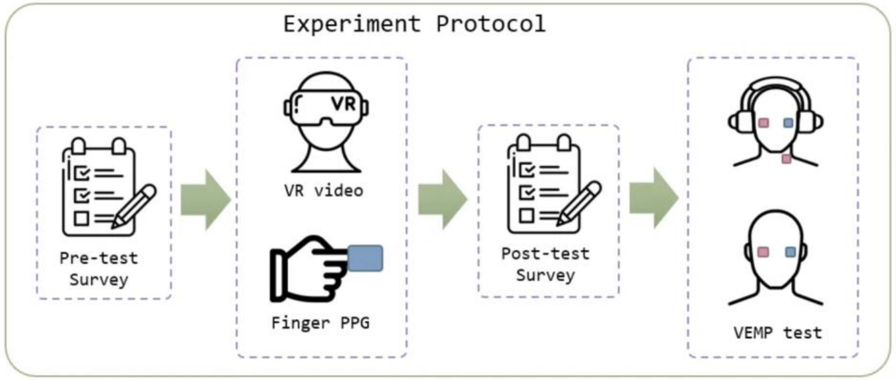


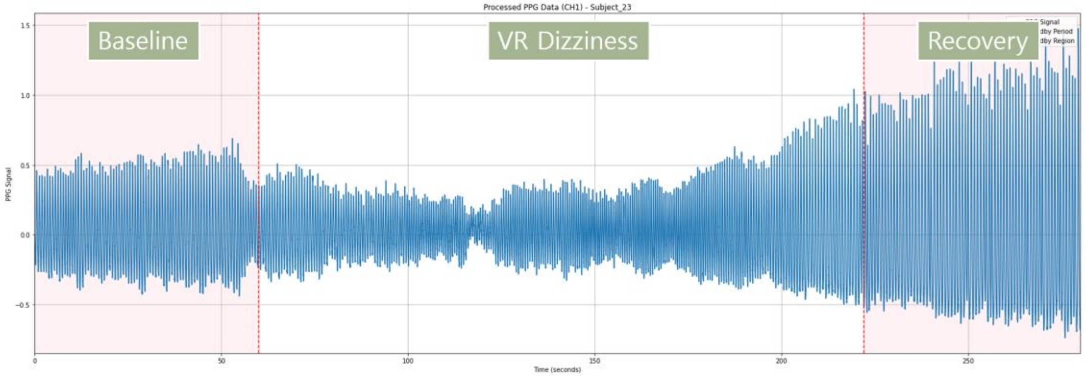
[Feature engineering]

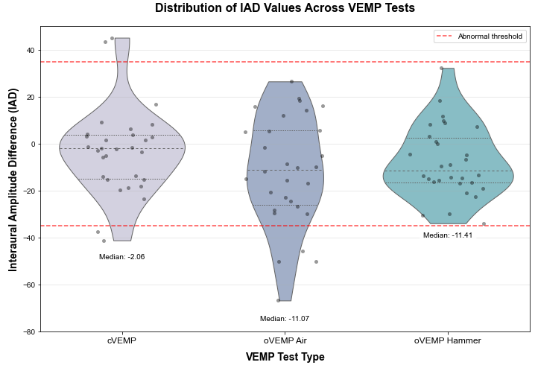


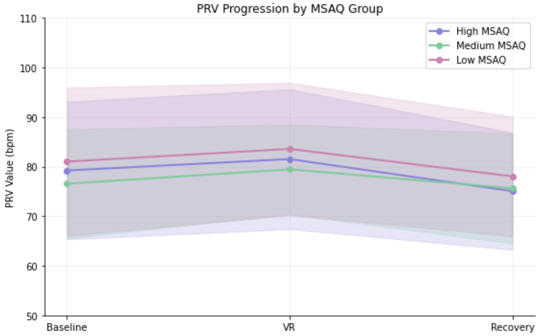
[Classification results]

Cybersickness Detection Research







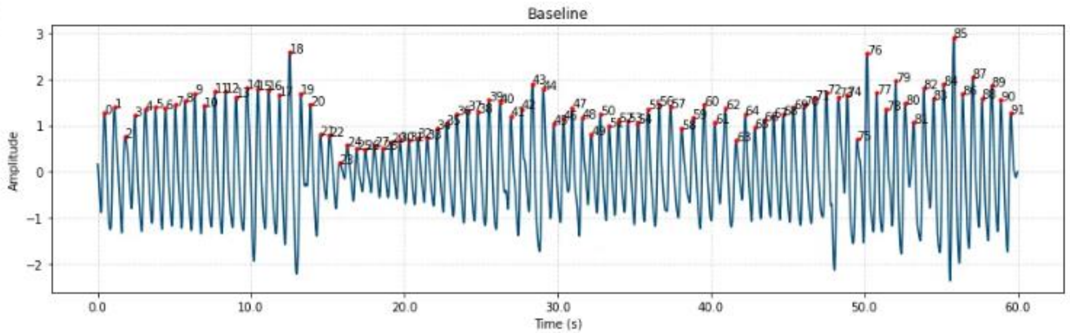


DEVELOPMENT OF A CYBER-SICKNESS FREE SYSTEM TO OVERCOME THE LIMITATION OF XR DIGITAL THERAPEUTICS

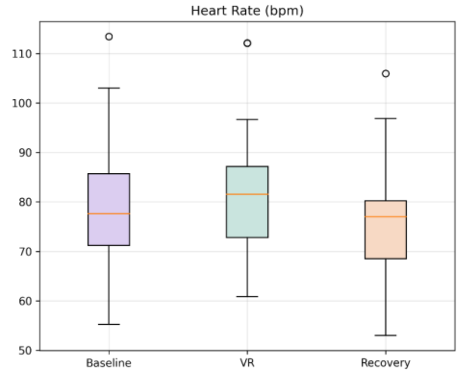
- **Objective:** To develop an objective measure of cyber-sickness in XR digital therapeutics by analyzing physiological responses during virtual reality exposure.
- **Methods:** Data analysis between non-invasive finger PPG-based monitoring and autonomic nervous system changes
  - Analyzed pulse rate variability (PRV) as potential objective indicator of cybersickness
  - Investigated correlation between PRV and VEMP
- **Results:**
  - Observed significant HR changes during VR (+3.67%) and recovery (-6.30%) phases
  - Found correlations between VR sickness severity and vestibular function (r=0.551, p=0.002)
  - Under review in Journal of Clinical Neurology (2025, SCIE Q2)
- **Skills developed:** Autonomic nervous system analysis, PPG signal processing, clinical correlation studies, VR-related biomarker research

Cybersickness Detection Research

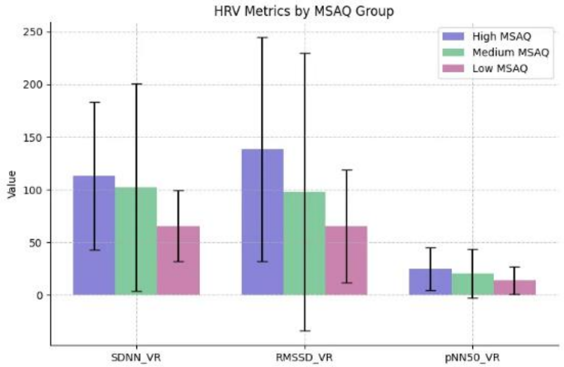
- **Reference Standard:** Used validated questionnaires (MSAQ, MSS-F, MSS-S, VR-SSS) and vestibular evoked myogenic potentials (VEMP) to assess motion sickness susceptibility in 30 participants
- **PPG Signal Processing:**
  - Recorded finger PPG at 2000 Hz during baseline (60s), VR exposure (162s), and recovery (60s)
  - Extracted heart rate variability parameters across experimental phases
  - Segmented data to analyze autonomic nervous system response patterns
- **Feature Analysis:**
  - Measured heart rate variability metrics (HR, SDNN, RMSSD) across all phases
  - Performed statistical comparisons between phases using paired t-tests
  - Grouped participants by motion sickness susceptibility (high, medium, low)
- **Correlation Framework:**
  - Analyzed relationships between HRV parameters and subjective VR sickness reports
  - Identified correlations between vestibular function and motion sickness susceptibility
  - Evaluated potential biomarkers through statistical correlation analysis (p<0.05)



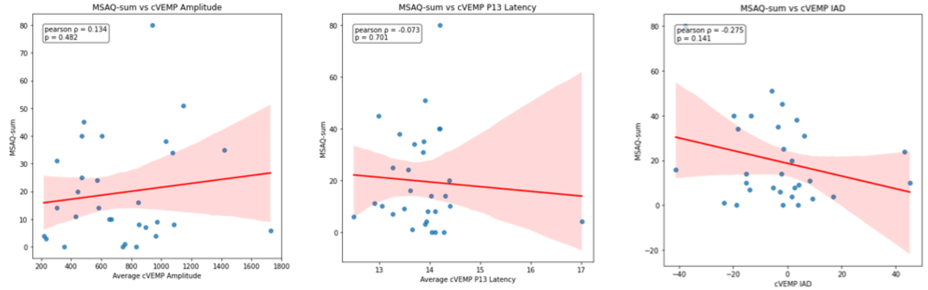
[PPG Signal Processing]



[Hear rate changes by phases]



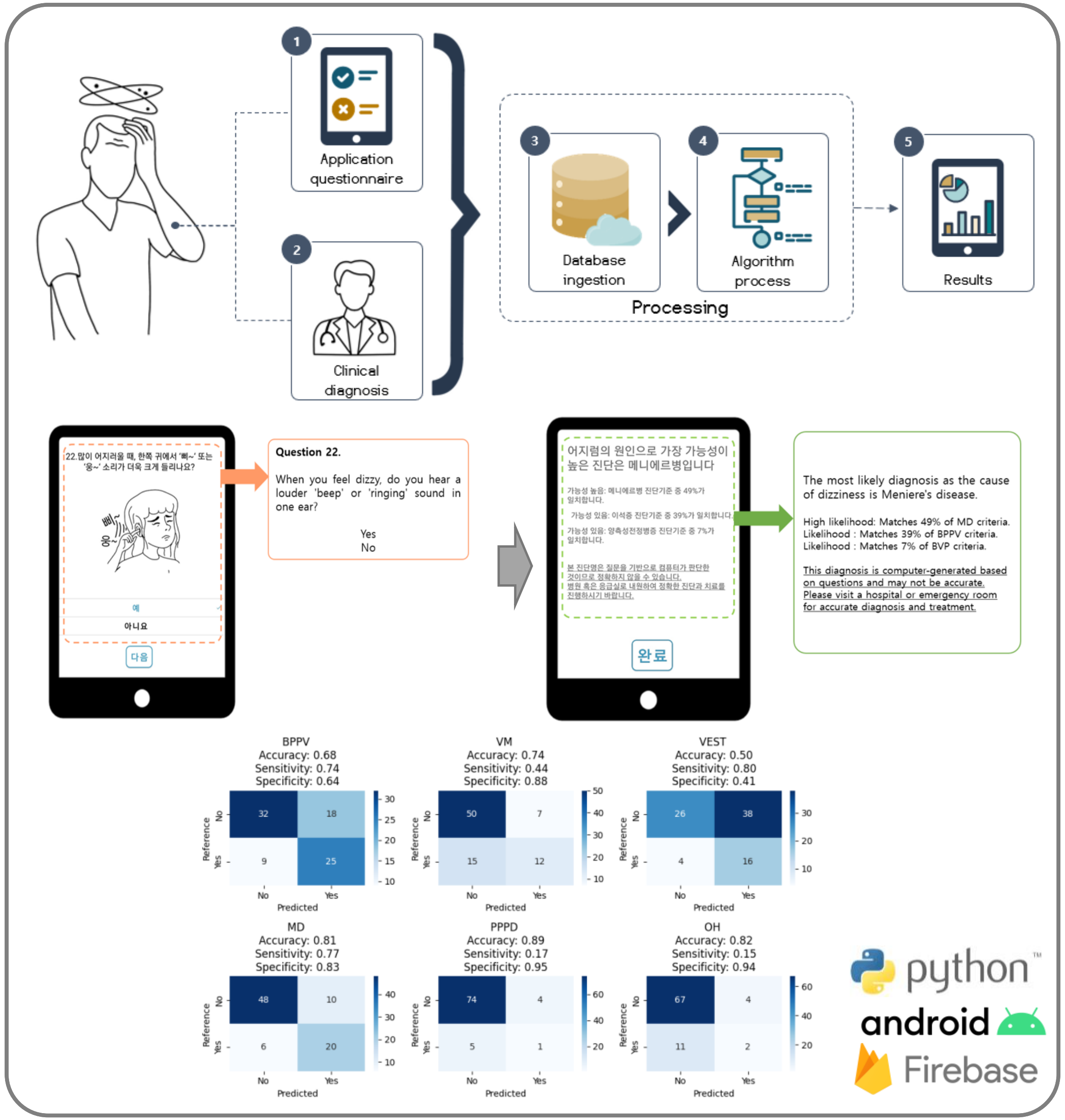
[HRV Metric by MSAQ group]



[Correlations between features]

Research on AI-based diagnostic tools

AI-Based Vestibular Disorder Diagnostic System



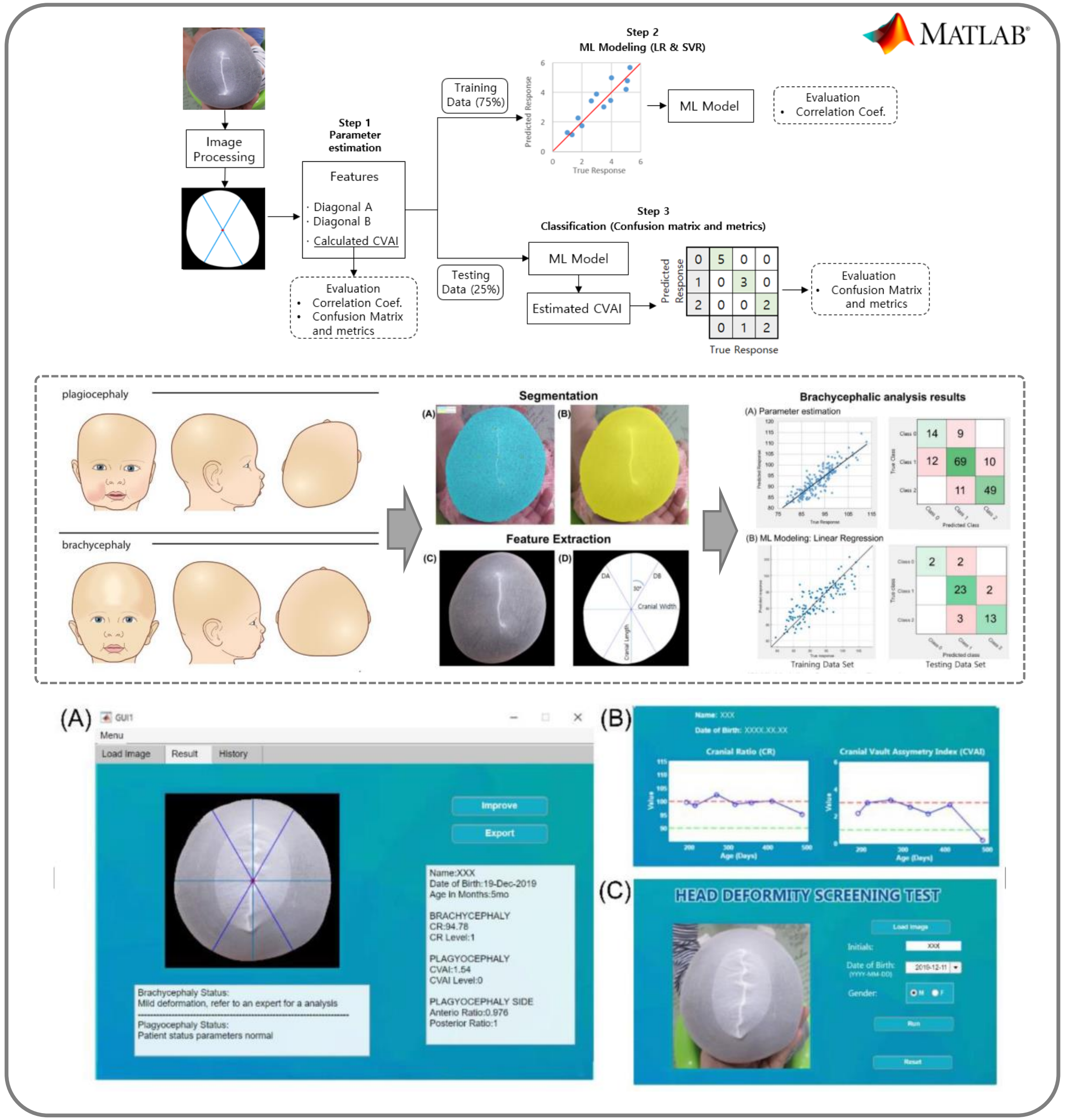
CLINICAL DECISION SUPPORT FOR VESTIBULAR DIAGNOSIS

- **Objective:** To develop a mobile application screening tool for six common vestibular disorders using rule-based & AI algorithms.
- **Methods:** Developed Flutter-based mobile questionnaire system in collaboration with ENT specialists
  - Implemented a 47-item questionnaire to analyze symptoms
  - Created AI models analyzing 3,679 patient records
- **Results:**
  - Achieved 82.1% diagnostic accuracy with rule-based algorithm (107 participants)
  - Improved to 86% classification accuracy with AI models
  - Patent application completed (KR10-2023-0055247)
  - ① Under review in Scientific Reports (2025, SCIE Q1)
  - **② Published in NPJ Digital Medicine (2025, SCIE Q1)**
- **Skills developed:** Android mobile application development, Flutter development, healthcare UI/UX design, clinical workflow implementation, diagnostic algorithm development, cross-platform mobile development, ML algorithms optimization

Cranial Deformity Screening Tool

TWO-DIMENSIONAL IMAGE-BASED SCREENING FOR POSITIONAL CRANIAL DEFORMITIES

- **Objective:** To develop an innovative, cost-effective screening method for infant cranial deformities using 2D vertex images and machine learning algorithms.
- **Methods:**
  - Created MATLAB UI with automated image processing pipeline
  - Implemented image processing and countour detection
  - Extracted accurate Cranial Ratio and Cranial Vault Asymmetry Index measurements
- **Results:**
  - Achieved strong correlation with gold-standard 3D scanner (0.85-0.89)
  - Published in Diagnostics (2020, SCIE Q1)
  - Korean patent registered (KR2377364)
- **Skills developed:** MATLAB programming, medical image processing, feature extraction algorithms, user interface design and statistical analysis

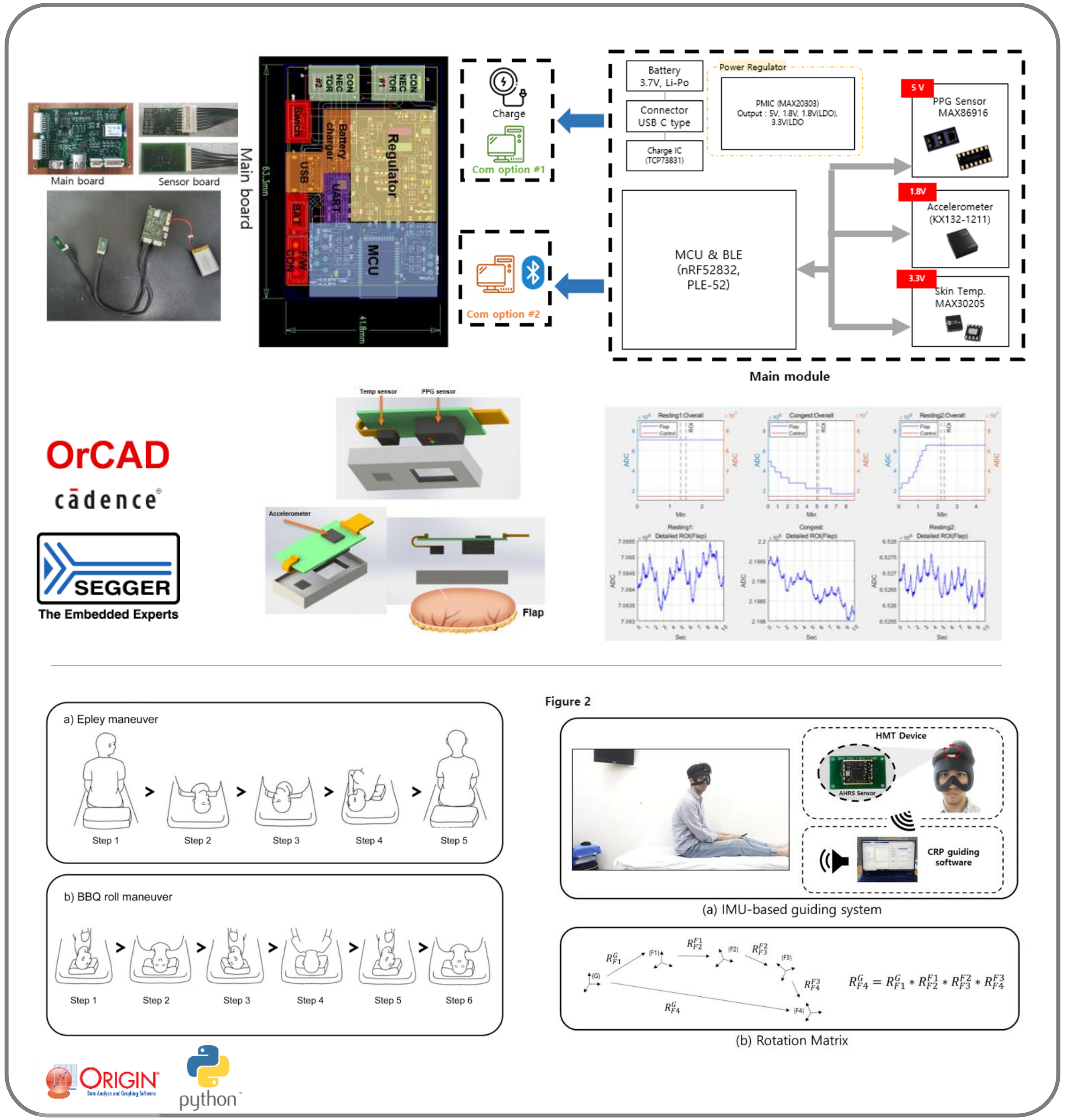


Research on healthcare wearable system

Research on healthcare wearable system

DEVELOPMENT OF ADVANCED MEDICAL SENSING SYSTEMS FOR SURGICAL MONITORING AND VESTIBULAR TREATMENT

- **Objective:** To develop advanced medical devices for healthcare applications, including a smart flap monitoring system using AIoT and an IMU sensor-based guiding system for BPPV treatment.
- **Methods:**
  - Designed wearable devices integrating multi-wavelength PPG sensors, skin temperature sensors, and IMU sensors
  - Developed dual-sensor configuration with reference monitoring for comparative tissue perfusion analysis
  - Created algorithms to detect ischemia/congestion from PPG signal components and guide head rotations for BPPV treatment
- **Results:**
  - Validated IMU-CRP system in clinical study (N=19) demonstrating improved BPPV treatment accuracy
  - Successfully implemented real-time monitoring with predictive analytics for flap procedures
  - Published findings in Scientific Reports (2023, SCIE Q1)
- **Skills developed:** Biomedical sensor integration, hardware design, Bluetooth communication, medical device certification protocols, IMU sensor signal processing, real-time monitoring systems, clinical validation methodologies



Potential Contributions to Samsung Health Research

Thank you

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Appendix

INDUSTRY EXPERIENCE

MedInTech:

- Designed and developed a real-time GPU monitoring web server
- MedInTech Gateway software for endoscopic video capture
- Designed and developed MD-SC-300 AI-powered cancer classification system

IDETECA S.R.L. :

- Evaluated medical device software/hardware components using specialized sensor technology
- Designed electrical/mechanical safety testing protocols compliant with IEC 60601 standards
- Verified sensor output signal stability/accuracy using oscilloscopes, data loggers, spectrum analyzers

IMEXCOMED S.R.L. :

- Performed preventive/corrective maintenance for endoscopy and surgical equipment
- Conducted component replacement, circuit testing, and on-site troubleshooting
- Provided installation services and advanced software support in hospital environments